

Wi-Fi

6.1 Introduction and history

Wireless is obviously no longer a stranger to most people. Even for someone unacquainted with the concept of RF transmissions or any mode of wireless communication, Wi-Fi would not be a new term. Almost every individual who owns a laptop would probably have been witness to a Wi-Fi signal once in a while. Even in a country like India where wireless zones aren't really as popular as in the west, there are several housing societies, corporate houses, and educational institutions who have access to Wi-Fi – your portal to wireless internet. Today, Wi-Fi is synonymous with WLAN (Wireless Local Area Networks) in most countries. Its uses are versatile and its technology quite established and prolific. Through the course of this chapter, we shall describe the history of Wi-Fi, the basic overview of its working, the devices that operate it, the features it offers to users, the potential it has, the security issues regarding Wi-Fi and how you can enable wireless networking on your computer.

History

Wi-Fi is part of radio technology, where data is transmitted using radio waves of high frequency. As mentioned before, there is a part of the electromagnetic spectrum which is unlicensed. Anyone can utilise that part of the spectrum for their purposes. This unlicensed spread spectrum was first available in the USA in 1985. The Federal Communications Commission laid down stringent regulations for usage of the spectrum which was later implemented in most other countries as well. The spectrum was intended for civilian usage, and is responsible for almost all free wireless technology currently available.

With the growth of Wi-Fi, the term is now synonymous with wireless LAN

The precursor to Wi-Fi was invented in 1991 by the NCR Corporation and AT&T in the Netherlands. Intentionally meant for cashier systems, the first wireless products were introduced